

Advanced AI-Driven Prediction of Pregnancy-Related Adverse Drug Reactions

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Cite This: *J. Chem. Inf. Model.* 2024, 64, 9286–9298



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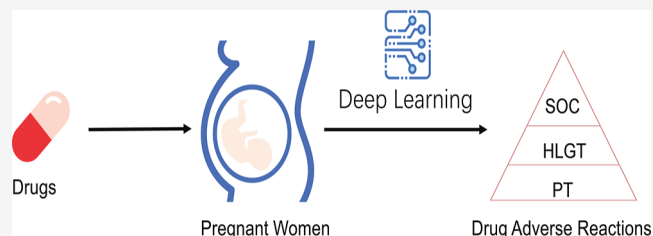


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ABSTRACT: Ensuring drug safety during pregnancy is critical due to the potential risks to both the mother and fetus. However, the exclusion of pregnant women from clinical trials complicates the assessment of adverse drug reactions (ADRs) in this population. This study aimed to develop and validate risk prediction models for pregnancy-related ADRs of drugs using advanced Machine Learning (ML) and Deep Learning (DL) techniques, leveraging real-world data from the FDA Adverse Event Reporting System. We explored three methods—Information Component, Reporting Odds Ratio, and 95% confidence interval of ROR—for classifying drugs into high-risk and low-risk categories. DL models, including Directed Message Passing Neural Networks (DMPNN), Graph Neural Networks, and Graph Convolutional Networks, were developed and compared to traditional ML models like Random Forest, Support Vector Machines, and XGBoost. Among these, the DMPNN model, which integrated molecular graph information and molecular descriptors, exhibited the highest predictive performance, particularly at the preferred term level. The model was validated against external data sets from SIDER and DailyMed, demonstrating strong generalizability. Additionally, the model was applied to assess the risk of 22 oral hypoglycemic drugs, and potential substructure alerts for pregnancy-related ADRs were identified. These findings suggest that the DMPNN model is a valuable tool for predicting ADRs in pregnant women, offering significant advancement in drug safety assessment and providing crucial insights for safer medication use during pregnancy.



INTRODUCTION

Medication use during pregnancy is widespread, with research showing that about 97% of pregnant women use at least one medication during their pregnancy and approximately 30% report using at least five medications.¹ Furthermore, the incidence of multiple medication use during pregnancy increased from 2.8% in 1999–2000 to 10.0% in 2015–2016.² Thoroughly evaluating the safety of medications for both the mother and fetus is essential. However, ethical and economic considerations typically exclude pregnant women from clinical trials, leading to a significant gap in clinical medication information for this group. In vitro and animal studies often fail to accurately reflect adverse drug reactions (ADRs) in pregnant women or fetuses, sometimes yielding false-negative results, as seen with thalidomide's severe fetal malformations. Conversely, false-positive results can limit medication use during pregnancy,³ hindering the treatment of pregnancy-related conditions. Furthermore, ADRs during pregnancy are primarily detected through postmarketing surveillance, which systematically monitors and assesses drug effects but often suffers from delays. Consequently, effectively evaluating and predicting the side effects of medications on pregnant women and fetuses remains a significant challenge in drug development and clinical practice.

Deep Learning (DL) and Machine Learning (ML) have rapidly been adopted in various fields, including healthcare, for

their ability to handle large data sets, identify patterns, and make predictions. These technologies show significant potential in disease diagnosis, drug discovery and design, personalized treatment, and patient care management.⁴ DL and ML significantly enhance our ability to detect and predict ADRs and toxicity by efficiently processing and analyzing diverse data sources, including electronic health records (EHRs), genomic data, and clinical trial reports. Despite these advances, research on ADRs specific to pregnant women remains limited. For instance, Boland et al.⁵ utilized EHRs and cohort study data to build logistic regression and random forest (RF) models to predict the FDA pregnancy categories (A, B, C, D, X) of drugs, assessing their potential harm to the fetus. Similarly, Shtar et al.⁶ used manually annotated teratogenic data to develop a multimodal model for predicting teratogenic risks, achieving notable predictive performance. However, the scarcity of ADR data for pregnant women has restricted these studies to modeling only the teratogenic effects

Received: September 11, 2024

Revised: November 19, 2024

Accepted: November 20, 2024

Published: November 29, 2024



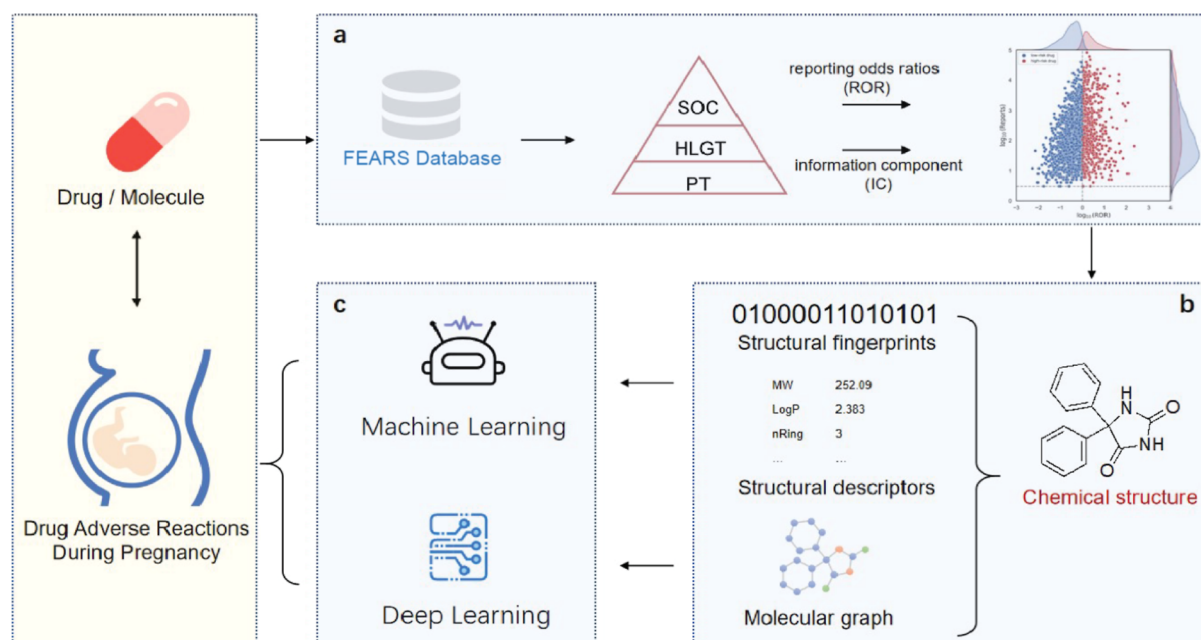


Figure 1. Workflow for data processing and modeling to predict adverse reactions during pregnancy.

of drugs, lacking a comprehensive prediction model for all ADRs during pregnancy. This gap leaves a critical question for pharmaceutical companies and healthcare providers: which medications, particularly investigational or newly marketed drugs, may cause adverse reactions in pregnant women?

Developing prediction models for ADRs during pregnancy presents several challenges: the lack of comprehensive data sets with known pregnancy-related ADR risks and control data, the selection and acquisition of relevant drug features, and the choice of appropriate modeling techniques. This study addressed these issues by extracting pregnancy-related ADR reports from the FAERS database, evaluating the impact of different data processing methods and comparing three methods for categorizing drugs into high-risk and low-risk groups. The study utilized 2D molecular descriptors, fingerprints [ECFP4, molecular access system (MACCS)], and molecular graphs, employing ML algorithms [support vector machines (SVM), XGBoost, and RF] and DL techniques [graph convolutional network (GCN), graph neural network (GNN), GAT, and DMPNN] to build models. To ensure robustness, the models were validated by using external data from the SIDER and DailyMed databases. This study aims to develop AI-based prediction models to identify pregnancy-related ADRs, and it offers a valuable tool for researchers, clinicians, and the pharmaceutical industry. This approach is intended to facilitate more informed decision-making in drug development and clinical practice.

METHODS

The workflow for developing an AI prediction model for identifying pregnancy-related ADRs involves several key steps: data collection and processing, molecular characterization computation, predictive model construction, and model evaluation. These steps are illustrated in Figure 1.

Data Collection. Data for this study were sourced from the FAERS database (<https://fis.fda.gov/extensions/FPD-QDE-FAERS/FPD-QDE-FAERS.html>), which classifies ADRs using the MedDRA standardized dictionary. This system

organizes data into hierarchical levels including System Organ Class (SOC), High-Level Group Term (HLGT), and Preferred Term (PT). To ensure a thorough analysis, all pregnancy-related adverse event reports were consolidated. After deduplication and retaining only the latest reports as part of the preprocessing of the FAERS database, we examined 50,231,908 adverse events reported between January 2004 and December 2023. During data processing, pregnancy-related PT adverse events categorized under the SOC level “Pregnancy, Puerperium, and Perinatal Conditions” were extracted and further classified into their corresponding HLGT categories. The associated drugs were identified, with their generic names retrieved from the DrugBank database (<https://go.drugbank.com/>). Various methods were employed to distinguish high-risk and low-risk drugs for pregnancy-related adverse reactions across these levels. At the PT level, we focused on PT classes with more than 200 drugs and a high-to-low-risk drug ratio within a factor of 2, excluding PT classes unrelated to diseases.

Additionally, drugs linked to pregnancy-related adverse reactions were collected from the SIDER database, and drug labels from the DailyMed database were thoroughly screened to identify those that could cause fetal malformations. All of the identified adverse reactions were classified at the PT level. These drugs formed an external test set, which was meticulously deduplicated from the modeling data—referred to as the unseen data set—to ensure the accuracy and reliability of the model evaluations.

All drugs in the data set were limited to small molecule drugs, with their molecular structures obtained from the DrugBank database and saved as Simplified Molecular Input Line Entry System (SMILES) strings. Drugs without available SMILES data were excluded. To ensure consistency and accuracy, molecular structures were standardized using the “wash” function of the Molecular Operating Environment software (version 2022.02, Chemical Computing Group, Montreal, QC, Canada). This process included removing compounds and salts, deprotonating strong acids and bases,

and adding explicit hydrogens to facilitate subsequent calculations of molecular descriptors and molecular graphs.

Methods for Classifying Drugs into High-Risk and Low-Risk. ADR signal detection in spontaneous reporting systems primarily includes Bayesian approaches, such as the Bayesian Confidence Propagation Neural Network (BCPNN), and frequency-based approaches, such as the reporting odds ratio (ROR).⁷ This study employed three methods to assess the risk of pregnancy-related ADRs for drugs.

Method 1 (Information Component Method). The Information Component (IC) is a metric used in Bayesian data mining methods such as BCPNN to detect ADR signals. It measures the deviation of the observed frequency of drug-ADR combinations from the expected frequency, based on prior probabilities and observed data.⁸ High-risk drugs are identified if $IC-2SD > 0$, $IC = E(IC_{ij})$, $SD = \sqrt{V(IC_{ij})}$. The $E(IC_{ij})$ and $V(IC_{ij})$ values are defined by the following formula

$$V(IC_{ij}) = \frac{\frac{N - C_{ij} + \gamma - \gamma_{ij}}{(C_{ij} + \gamma_{ij})(1 + N + \gamma)} + \frac{N - C_i + \alpha - \alpha_i}{(C_i + \alpha_i)(1 + N + \alpha)} + \frac{N - C_j + \beta - \beta_j}{(C_j + \beta_j)(1 + N + \beta)}}{(\log_2)^2} \quad (1)$$

$$E(IC_{ij}) = \log_2 \frac{(C_{ij} + \gamma_{ij})(N + \alpha)(N + \beta)}{(N + \gamma)(C_i + \alpha_i)(C_j + \beta_j)} \quad (2)$$

$$\gamma = \gamma_{ij} \times \frac{(N + \alpha)(N + \beta)}{(C_i + \alpha_i)(C_j + \beta_j)} \quad (3)$$

$$C_{ij} = A \quad (4)$$

$$C_i = A + B \quad (5)$$

$$C_j = A + C \quad (6)$$

$$N = A + B + C + D \quad (7)$$

Here, γ_{ij} , α_i , β_j , α , and β are constants, and A , B , C , and D represent the number of reports in the FAERS database with different treatment and outcome combinations: A is the number of reports where individuals received the target drug and experienced a specific adverse event (SOC, HLG, or PT); B is the number where individuals received the target drug but did not experience the adverse event; C is the number where individuals did not receive the target drug but experienced the adverse event; D is the number where individuals did not receive the target drug and did not experience the adverse event.

Method 2 (ROR + CI Method). The ROR is used in pharmacovigilance to measure the association between a drug and an ADR. It compares the odds of a specific ADR being reported for a particular drug with the odds of the same ADR being reported for all other drugs in the database.⁹ A drug is considered high-risk if the ROR value and the lower limit of 95% confidence interval (95% CI) of ROR exceed 1. The ROR and 95% CI are calculated as follows

$$ROR = \frac{A/B}{C/D} \quad (8)$$

$$SE = \sqrt{1/A + 1/B + 1/C + 1/D} \quad (9)$$

$$95\% \text{ CI} = e^{\ln ROR \pm 1.95 \times SE(\ln ROR)} \quad (10)$$

Method 3 (ROR Method). The ROR, which measures the odds of reporting a specific adverse event for a particular drug relative to all other drugs, showed that a $ROR > 1$ indicates a higher likelihood of reporting the adverse reaction within the group of interest.¹⁰ The ROR method classifies drugs for pregnancy-related adverse reactions based solely on the ROR value. Drugs with $ROR > 1$ are deemed high risk, while those with $ROR < 1$ are considered low risk.

Molecular Structure Characterization. Traditional ML methods represent molecules using molecular descriptors and fingerprints, while DL methods utilize molecular graphs. In our study, we characterized drug structures using 2D molecular descriptors, two types of fingerprints (ECFP4 and MACCS), and molecular graphs, building on methodologies previously reported by our group.¹¹

2D Molecular Descriptors. We calculated 208 2D molecular descriptors using the `rdkit.Chem.Descriptors` function from the RDKit package (version 2023.9.6). These descriptors include molecular weight, partition coefficient ($\log P$ using Wildman and Crippen methods), hydrogen bond donors and acceptors (NumHDonors and NumHAcceptors), topology (TPSA, kappa1–3, BertzCT), composition (NumRings, NumAromaticRings), electrotopological state (Estate), etc. This descriptor set is referred to as Des. A complete list is available in the RDKit documentation (<https://www.rdkit.org/docs/GettingStartedInPython.html#list-of-available-descriptors>).

Fingerprints. The MACCS fingerprint encodes the presence (1) or absence (0) of predefined chemical features in a 166 bit binary string. The Extended Connectivity Fingerprint (ECFP4) addresses the isomorphism problem by breaking molecules into circular fragments and encoding each atom by its type. The ECFP4 fingerprint has a radius of 2 and is typically represented as a 1024 bit binary string. Both MACCS and ECFP4 fingerprints were computed using the RDKit package (version 2023.9.6).

Molecular Graphs. Molecular graph-based methods use graph convolutional operations to learn molecular representations. In a molecular graph, connectivity between atoms is represented by a graph $[G = (V, E)]$, where nodes (V) represent atoms with feature vectors (X_v) and edges (E) represent bonds with feature vectors (E_{km}), where km denotes the bond between atoms k and m . A molecule can thus be viewed as a graph composed of nodes (atoms) and edges (bonds). DL methods utilize convolutions on molecular graphs to create representations of a drug's structure. These atomic representations are then used during the readout stage to predict the drug's molecular properties.¹¹

Model Construction. The performance of various traditional ML and DL algorithms on pregnancy ADR models was examined. Descriptor-based models were built by using three ML algorithms: SVM, XGBoost, and RF. Additionally, DL models were constructed using four graph-based methods: GCN, GNN, graph attention network (GAT), and directed message passing neural network (DMPNN). Modeling was performed at three levels: SOC, HLG, and PT.

Descriptor-Based Models. These models utilized two types of fingerprints and one set of descriptors generated using RDKit: ECFP4, MACCS, and Molecular Descriptors.¹² The SVM model employed the radial basis function kernel, optimizing hyperparameters C (0.1 to 100) and γ (0 to 0.2). For XGBoost, optimized hyperparameters included

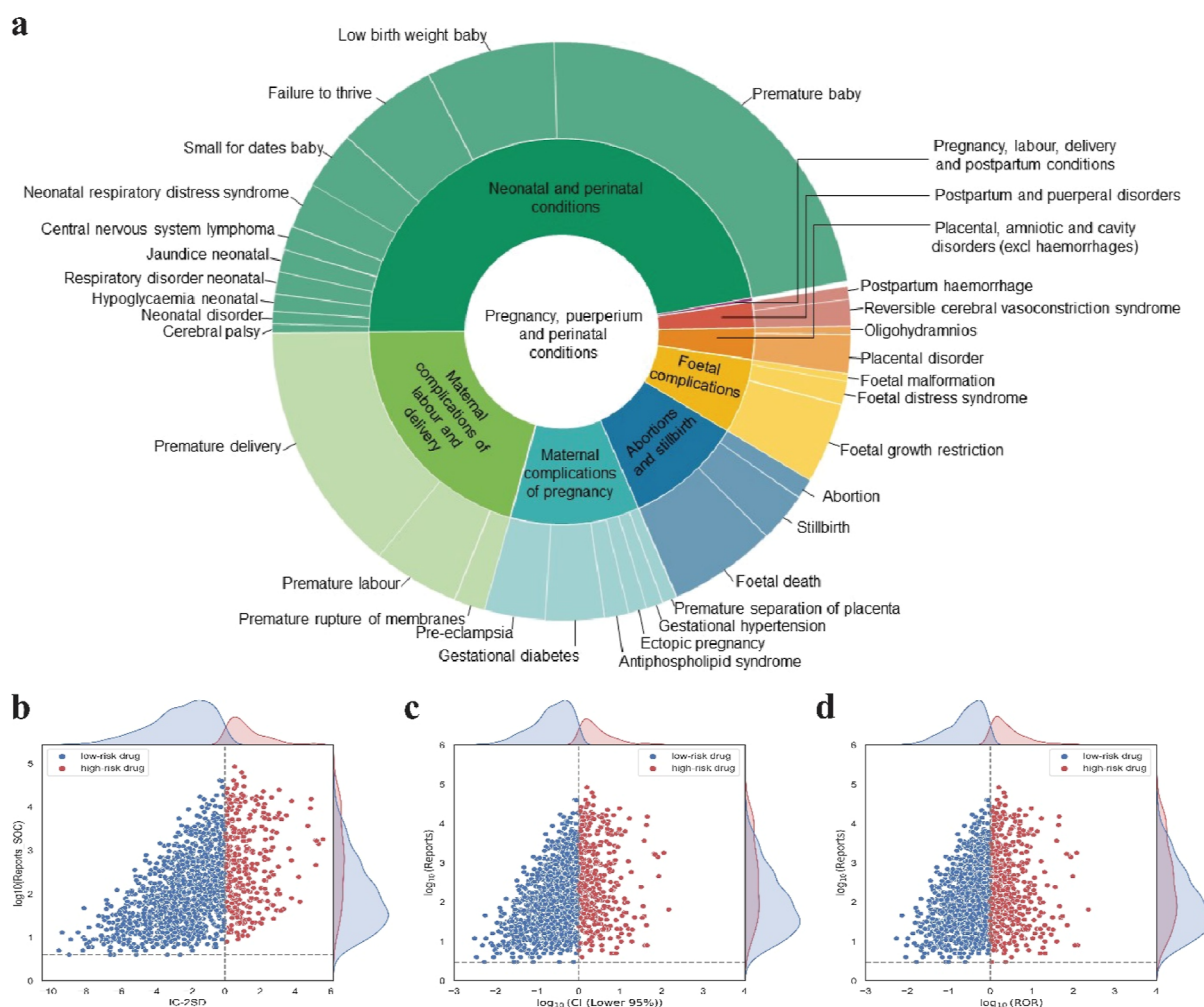


Figure 2. Reports and distribution of pregnancy-related ADRs and drugs. (a) The sunburst chart illustrates the proportion of pregnancy-related ADRs at each hierarchical level, from inner to outer: SOC, HLGT, and PT. (b) Scatter plots depict signal detection metrics for Method 1 (IC method), specifically the IC-2SD value, against the number of adverse reaction reports at the SOC level. (c) Scatter plots depict signal detection metrics for Method 2 (ROR + CI method), specifically the lower limit of the ROR's 95% confidence interval, against the number of adverse reaction reports at the SOC level. (d) Scatter plots depict signal detection metrics for Method 3 (ROR method), specifically the ROR value, against the number of adverse reaction reports at the SOC level. The dotted horizontal line indicates where the number of reports equals 3.

learning_rate (0.01 to 0.2), gamma (0 to 0.2), min_child_weight (1 to 6), subsample (0.7 to 1.0), collaboration (3 to 10), and $n_{\text{estimators}}$ (50 to 1000). For RF, optimized hyperparameters were $n_{\text{estimators}}$ (10 to 500), max_depth (3 to 12), min_samples_leaf (1 to 50), min_impurity_decrease (0 to 0.01), and max_features ("sqrt", "log2", 0.7 to 0.9).¹¹

Graph-Based Models. GNN models update randomly initialized atom vectors to derive molecular vectors by leveraging the molecular graph structure and then predict molecular properties through backpropagation.¹³ To optimize the models, hyperparameters such as learning rate (1×10^{-5} to 1×10^{-3}), dimension (50 to 200), number of hidden layers (2 to 5), output layers (2 to 5), and batch size (16 to 24) were fine-tuned.

GCN employed a scalable semisupervised learning method for graph-structured data. It utilized a localized first-order approximation of spectral graph convolutions to operate directly on graphs.¹⁴ Hyperparameters for the GCN model were optimized, including hidden size for GNN layers (32 to 256), hidden size for the multilayer perceptron (MLP) predictor (16 to 1024), number of GCN layers (1 to 5), use

of residual connections for each GCN layer (true/false), application of batch normalization (true/false), and dropout rate.

GAT updated node representations using an attention mechanism built on the GCN framework.¹⁵ For the GAT model, hyperparameters such as hidden size for each attention head in GNN layers (32 to 256), number of attention heads in each GNN layer (4 to 8), slope for negative values in LeakyReLU (0 to 1), hidden size for the MLP predictor (16 to 256), number of GNN layers (1 to 5), use of residual connections for each GAT layer (true/false), and dropout rate were optimized.

DMPNN learned molecular encoding through bond-centric convolution, which avoids unnecessary cycles during message passing.^{16–18} studies have demonstrated that combining DMPNN with external features enhances performance, as descriptors provide global information while DMPNN offers local information.^{19–22} In this study, the DMPNN model was enhanced by integrating additional vectors of 2D molecular descriptors or ECFP4 fingerprints at the molecular level, termed DMPNN-Des and DMPNN-ECFP4, respectively. Hyperparameters such as the number of messages passing

Table 1. Number of Drugs with Pregnancy-Related Adverse Reactions Classified by MedDRA Categories Using Different Signal Detection Methods

MedDRA level	pregnancy ADR term	abbreviation	method1 (IC method)			method 2 (ROR + CI method)			method 3 (ROR method)		
			low-risk	high-risk	total	low-risk	high-risk	total	low-risk	high-risk	total
SOC	pregnancy, puerperium and perinatal conditions	Class0	1191	334	525	1106	414	520	1035	512	1547
HLGT	abortions and stillbirth	Class1	877	125	1002	810	188	998	695	247	942
	fetal complications	Class2	688	169	857	594	258	852	434	324	758
	maternal complications of labor and delivery	Class3	818	107	925	757	164	921	604	229	833
	maternal complications of pregnancy	Class4	767	115	882	678	201	879	471	272	743
	neonatal and perinatal conditions	Class5	984	281	1265	879	385	1264	719	447	1166
	placental, amniotic and cavity disorders (excl hemorrhages)	Class6	473	110	583	383	194	577	256	257	513
	postpartum and puerperal disorders	Class7	517	95	612	412	193	605	247	291	538
	pregnancy, labor, delivery, and postpartum conditions	Class8	1146	299	1445	1040	405	1445	924	516	1440
PT	premature baby	Class1	665	188	853	553	299	852	498	389	887
	premature delivery	Class2	557	148	705	466	233	699	399	326	725
	failure to thrive	Class3	528	106	634	384	243	627	249	400	649
	low birth weight baby	Class4	420	159	579	311	266	577	259	343	602
	fetal growth restriction	Class5	423	113	536	330	210	540	258	293	551
	premature labor	Class6	421	88	509	338	171	509	275	250	525
	pre-eclampsia	Class7	387	75	462	316	149	465	244	225	469
	fetal death	Class8	339	100	439	261	176	437	200	260	460
	neonatal respiratory distress syndrome	Class9	341	92	433	246	186	432	190	258	448
	gestational diabetes	Class10	363	74	437	290	142	432	247	199	446
	small for dates baby	Class11	334	81	415	265	148	413	223	204	427
	oligohydramnios	Class12	296	80	376	229	147	376	183	205	388
	stillbirth	Class13	306	55	361	264	99	363	236	147	383
	premature rupture of membranes	Class14	287	66	353	205	148	353	163	201	364
	fetal distress syndrome	Class15	264	51	315	206	107	313	149	178	327
	antiphospholipid syndrome	Class16	278	41	319	213	102	315	149	176	325
	abortion	Class17	290	28	318	244	72	316	203	122	325
	reversible cerebral vasoconstriction syndrome	Class18	223	59	282	150	132	282	105	180	285
	hypoglycemia neonatal	Class19	229	44	273	170	103	273	121	158	279
	ectopic pregnancy	Class20	248	20	268	214	54	268	178	97	275
	jaundice neonatal	Class21	232	39	271	186	84	270	132	142	274
	gestational hypertension	Class22	225	34	259	144	116	260	98	165	263
	neonatal disorder	Class23	218	38	256	162	93	255	120	141	261
	postpartum hemorrhage	Class24	217	38	255	151	103	254	105	152	257
	premature separation of placenta	Class25	202	38	240	166	71	237	121	125	246
	central nervous system lymphoma	Class26	174	44	218	138	76	214	115	106	221
	cerebral palsy	Class27	198	10	208	154	55	209	114	95	209
	respiratory disorder neonatal	Class28	145	53	198	113	86	199	88	116	204
	placental disorder	Class29	172	29	201	126	74	200	99	104	203
	fetal malformation	Class30	105	18	123	72	49	121	54	77	131

steps, number of layers in the feedforward neural network (FFN) model, dropout probability (0.0 to 1.0) in the MPNN and FFN layers, hidden dimension in the message passing step, and hidden dimension in the FFN model were optimized following the Chemprop 2.0.4 documentation.

The three descriptor-based models (XGBoost, RF, and SVM) were implemented in Python 3.8 using the scikit-learn package (version 0.20.1). Three graph-based models (GCN, GNN, GAT) were implemented using the Deep Graph Library package (version 0.4.1) with PyTorch as the backend.²³ The DMPNN model was implemented using the chemprop package (version 1.0).²⁴ All DL models were trained using the Adam optimizer, with hyperparameters fine-tuned through Bayesian optimization.²⁵

Model Evaluation. The data set was randomly divided into training, validation, and test sets in an 8:1:1 ratio. To evaluate the robustness and stability of the models, 5-fold cross-validation was employed. Specifically, the data set was divided into five parts; in each iteration, four parts were used for model training and one part for validation, repeated five times. The final model performance was evaluated by using the test set. Model performances were assessed by using six basic statistical metrics: area under the curve (AUC), accuracy (ACC), specificity (SP), sensitivity (SE), and F1 score. ACC indicates the model's overall prediction accuracy across the entire data set. SP measures the model's ability to identify low-risk drugs, with higher values indicating better performance in this aspect. SE measures the model's ability to identify high-

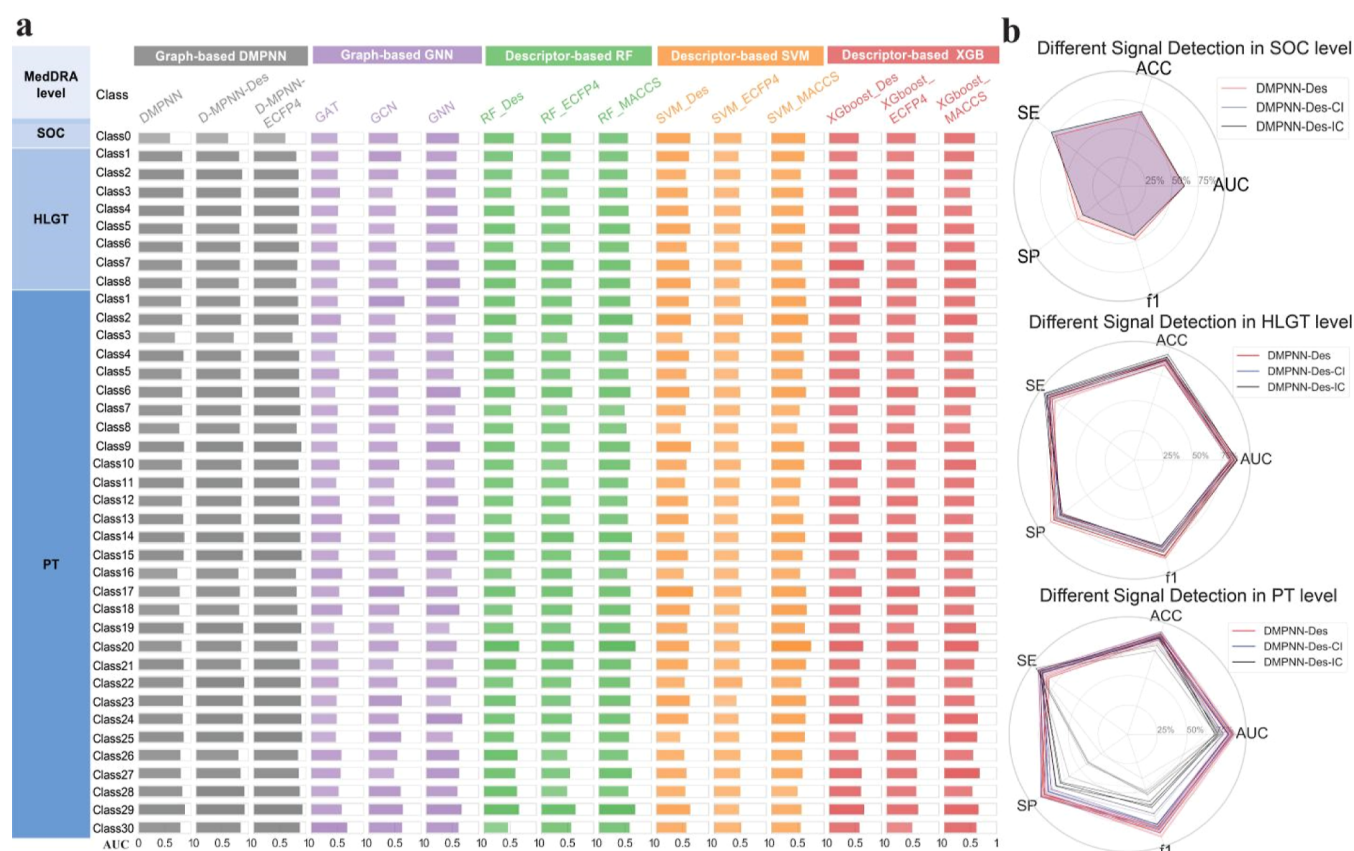


Figure 3. Metrics for ML and DL models predicting pregnancy-related adverse reaction risks. (a) AUC values pregnancy-related adverse reaction risk prediction models. The highest accuracy was achieved by the DMPNN, DMPNN-ECFP4, and DMPNN-Des models. Data from Method 3 (ROR method) were used to construct different models. (b) Performance of the DMPNN-Des model at the SOC, HLGT, and PT levels using data sets from three adverse reaction signal detection methods. The DMPNN-Des-IC Model was constructed using data sets from the adverse reaction detection Method 1 (IC method), DMPNN-Des-CI was Method 2 (ROR + CI method), and DMPNN-Des was Method 3 (ROR method).

risk drugs, with higher values indicating better performance in this aspect. The F1 score is the weighted average of the precision and recall, providing a comprehensive performance metric that accounts for both precision and recall. The calculation formulas for the evaluation metrics are as follows

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

$$SE = \frac{TP}{TP + FN} \quad (12)$$

$$SP = \frac{TN}{TN + FP} \quad (13)$$

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (14)$$

where TP, FP, TN, and FN represent the numbers of true positives, false positives, true negatives, and false negatives, respectively.

Analysis of Substructure Alert. To explore the role of substructures in pregnancy adverse reactions and guide the screening of pregnancy ADR risks during drug development, we used the PySmash toolkit developed by our research group.²⁶ Three substructure generation algorithms were employed for fragment extraction: functional, circular, and path fragments. For circular fragments, the minimum radius was set to 1 and the maximum radius was set to 4. For path

fragments, the minimum path length was set to 1 and the maximum path length was set to 7. To ensure the effectiveness of substructure extraction, the Bonferroni correction method was applied. The minimum number of compounds containing alert substructures was set to 5, with a minimum accuracy of 0.7 and a *P*-value threshold of 0.05.

$$\text{accuracy} = \frac{m_s}{n_s} \quad (15)$$

$$P_{\text{value}} = \sum_{i=ms}^{ns} \frac{n_s!}{i!(n_s - i)!} \left(\frac{m}{n}\right)^i \left(1 - \frac{m}{n}\right)^{ns-i} \quad (16)$$

where *n* is the total number of drugs in the data set, *m* is the number of drugs with positive data, *n_s* is the number of drugs containing a given substructure, and *m_s* is the number of drugs with positive data containing the given substructure.

RESULTS

Data Collection. A total of 487,554 pregnancy-related adverse event reports involving 2559 drugs were collected from the FAERS database and classified according to SOC, HLGT, and PT. The most frequently reported conditions were “Neonatal and perinatal conditions” at the HLGT level and “Premature baby” at the PT level (Figure 2a). Three adverse reaction signal detection methods were used to distinguish between low-risk and high-risk drugs, and the distribution of drugs at the SOC level are shown in Figure 2b–d. After the

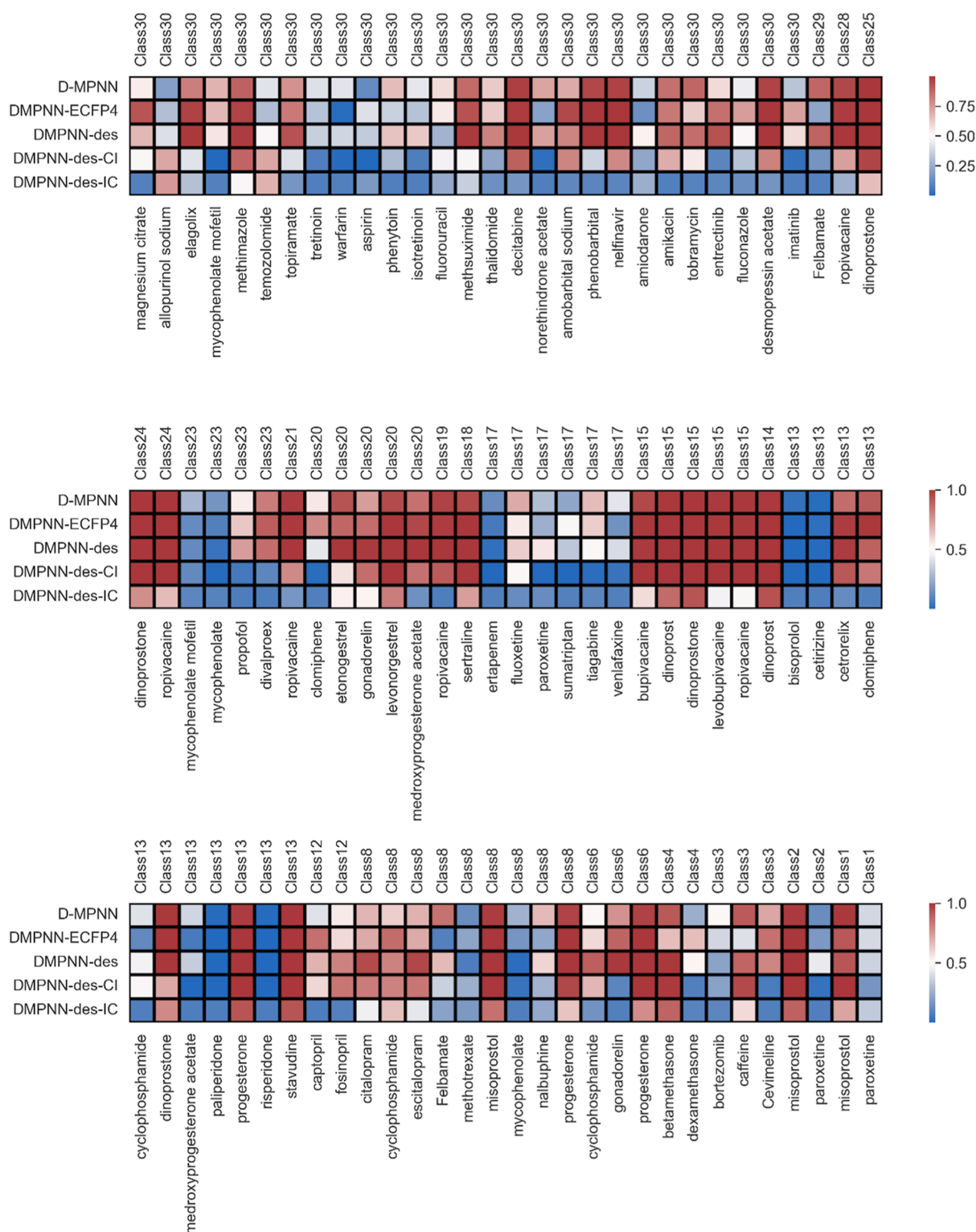


Figure 4. Results of DMPNN model predictions for high-risk pregnancy drugs from SIDER and DailyMed. The models include DMPNN, DMPNN-ECFP4, and DMPNN-Des, built using data from Method 3 (ROR method), and DMPNN-Des-IC and DMPNN-Des-CI, built using data from Method 1 (IC method) and Method 2 (ROR + CI method), respectively. The color gradient in the heatmaps represents prediction accuracy, with red indicating higher accuracy and blue indicating lower accuracy. Predictions greater than 0.5 (shades of red) are considered correct.

inclusion of drugs and adverse reactions with fewer than three reports, the number of low-risk and high-risk drugs at the SOC, HLGT, and PT levels is shown in Table 1. This includes 1 SOC-level pregnancy-related adverse reaction, 8 HLGT-level adverse reactions, and 30 PT-level adverse reactions. The data

set identified by Method 3 (ROR method) was more balanced compared to those identified by Method 1 (IC method) and Method 2 (ROR + CI method). Meanwhile, the high-risk drug data set obtained by Method 3 (ROR method) at the SOC, HLGT, and PT levels largely encompasses the high-risk drugs

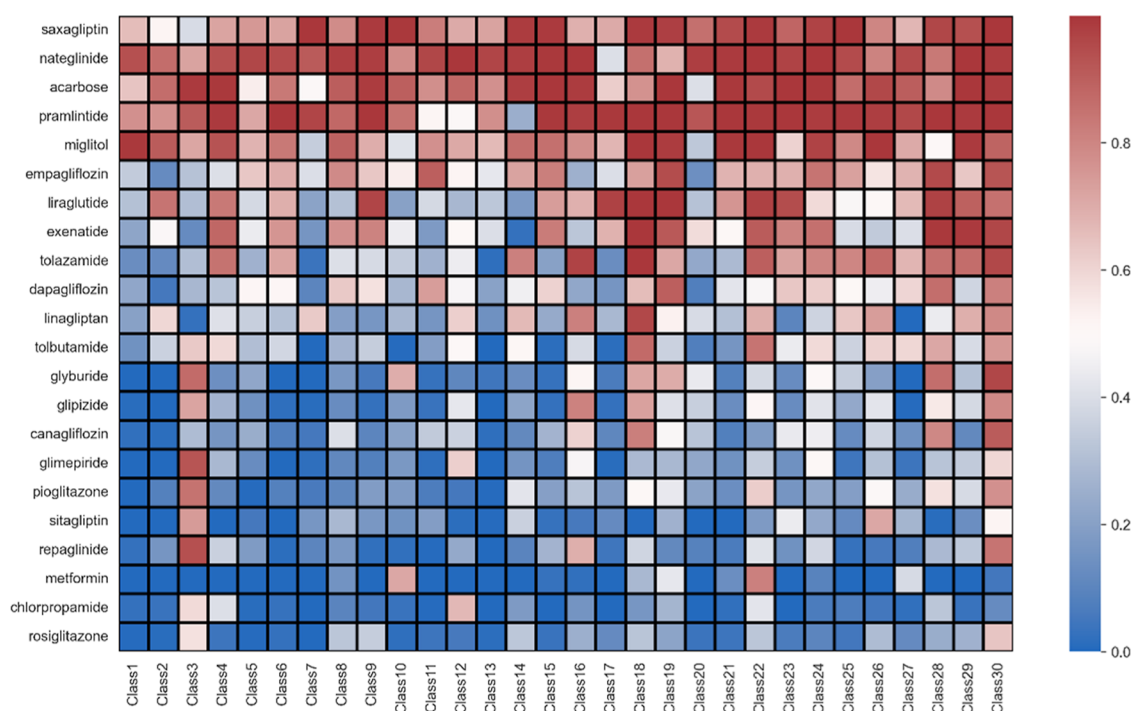


Figure 5. Predicted pregnancy-related adverse reaction risks of oral hypoglycemic drugs using the DMPNN-Des model. The heatmap displays the risk predictions, with blue indicating low risk and red indicating high risk.

identified by the other two methods. Similarly, the low-risk drugs identified by Method 1 (IC method) include those identified by the other two methods (Figure S1). These drugs were then used for model construction (Table S1).

Model Predictive Performance. In this study, we developed three traditional descriptor-based ML models and four graph-based DL models to identify the risk of pregnancy-related ADRs at the SOC, HLGT, and PT levels. The models' performance was evaluated using 5-fold cross-validation.

Among the models, the three DMPNN models, including the DMPNN model using only molecular graph information (DMPNN), the DMPNN model incorporating ECFP4 molecular fingerprint information (DMPNN-ECFP4), and the DMPNN model incorporating molecular descriptor information (DMPNN-Des), exhibited consistently high AUC values across most classes within HLGT and PT levels, indicating robust performance in predicting adverse reactions (Figure 3a, Table S2). Specifically, the AUC at the HLGT level was consistently above 0.8, ranging from 0.82 to 0.88. At the PT level, the DMPNN model showed AUCs below 0.8 for five adverse reactions (ranging from 0.70 to 0.79), whereas the DMPNN-ECFP4 and DMPNN-Des models each had AUCs below 0.8 for only one adverse reaction (Class 3), with values of 0.75 and 0.72, respectively. In contrast, the predictive performance of all three DMPNN models at the SOC level was lower with AUCs ranging from 0.61 to 0.62. Additionally, none of the other DL or ML models achieved an AUC of 0.8 or higher.

Models were also constructed using data sets from three adverse reaction detection methods: Method 1 (IC, DMPNN-Des-IC), Method 2 (ROR + CI, DMPNN-Des-CI), and Method 3 (ROR, DMPNN-Des). The performance of these models was evaluated. The results showed no significant difference between the models (DMPNN-Des, DMPNN-Des-CI, and DMPNN-Des-IC) at the SOC and HLGT levels

(Figure 3b). However, at the PT level, the DMPNN-Des model built using the data set from Method 3 demonstrated the best performance (Figure 3b). It is noteworthy that the model exhibited the highest performance at the PT level, followed by the HLGT level, while the predictive accuracy was the lowest at the SOC level.

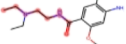
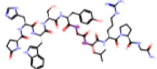
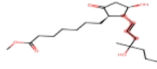
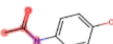
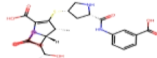
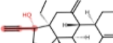
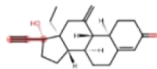
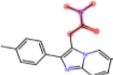
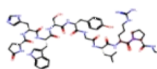
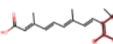
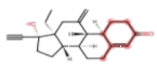
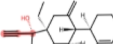
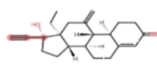
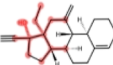
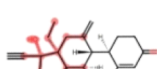
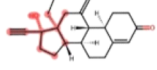
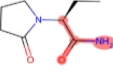
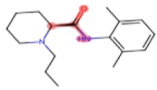
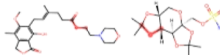
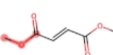
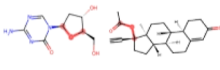
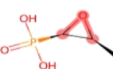
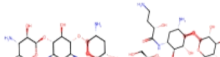
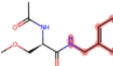
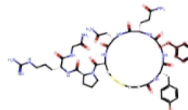
Performance Evaluation on the Unseen Data Set.

Using the SIDER database, we identified 41 drugs associated with 20 PT-level pregnancy-related adverse reactions. Additionally, screening drug labels from the DailyMed database revealed 27 drugs with the risk of causing fetal malformations. Thus, a total of 68 drugs and 21 PT-level adverse reactions were used as the unseen data set to validate the five DMPNN models.

The validation results indicated that models (DMPNN, DMPNN-ECFP4, DMPNN-Des) built using the data set identified by Method 3 (ROR method) outperformed those using Method 1 (DMPNN-Des-IC) or Method 2 (DMPNN-Des-CI). The descriptor-integrated DMPNN model (DMPNN-Des) showed the best performance. The accuracies for the DMPNN, DMPNN-ECFP4, DMPNN-Des, DMPNN-Des-IC, and DMPNN-Des-CI models were 69.23%, 67.03%, 72.53%, 49.45%, and 27.47%, respectively (Figure 4). These results demonstrate that the DMPNN-Des model constructed using Method 3 (ROR method) for identifying high-risk drugs had the best performance. In addition, a total of 21 adverse reactions at the PT level were predicted. Among these, Classes 13 and 17 consistently showed lower predictive accuracy across all models. Even with the best-performing DMPNN-Des model, the prediction accuracies for these classes were limited to 46% (4/11) and 33% (2/6), respectively.

Gestational Diabetes Mellitus and Drug Risk Assessment. Although oral hypoglycemic agents have not been approved by the FDA for the treatment of GDM, their use is increasing among pregnant women with GDM. Assessing the

Table 2. Pregnancy Risk Substructures of Drugs from FAERS and SIDER/DailyMed Database

PT		SMARTS		Pvalue	Accuracy	FAERS		SIDER or DailyMed	
Name	Class					Drug name	Structure	Drug name	Structure
Premature labour	Class6	CCNCCNC=O N, N'-Methylenediacetamide		0.041	0.76	metoclopramide		gonadorelin	
Foetal death	Class8	CC=CC 1-Butene		0.032	1.00	dimethyl fumarate		misoprostol	
Abortion	Class17	CC(N)=O Acetamide		0.012	0.88	acetaminophen		ertapenem	
		C#CC Propyne		0.001	1.00	desogestrel		etonogestrel	
Ectopic pregnancy	Class20	CN(C)C(C)=O N,N-Dimethylacetamide		0.019	0.78	zolpidem		gonadorelin	
		CC1=CCCCC1 Cyclohexene		0.010	0.80	tretinoin		etonogestrel	
		[*]-[C]#[CH] Terminal Alkyne		0.005	1.00	desogestrel		etonogestrel	
		[*]-[CH2]-[C](-[CH2]-[CH3])(-[CH](-[*])-[*H])(-[C](-[*])(-[*H])([*]) Isopentyl Group		0.041	1.00	desogestrel		etonogestrel	
		[*]-[CH2]-[C](-[OH])(-[C]#[CH])(-[C](-[*])(-[*H])([*]) 3-Hydroxypropynyl		0.041	1.00	desogestrel		etonogestrel	
Respiratory disorder neonatal	Class28	CC(N)=O Acetamide		0.019	1.00	levetiracetam		ropivacaine	
Foetal malformation	Class30								
		COC Dimethyl Ether		0.016	0.84	fosfomycin, dimethyl fumarate		mycophenolate mofetil, topiramate, decitabine, norethindrone acetate, amikacin, tobramycin, entrectinib	
									
		[*]-[CH2]-[c]1:[cH]:[cH]:[cH]:[cH]:[cH]:[cH]:1 Benzyl Group		0.046	1.00	lacosamide		desmopressin acetate	

risks of these drugs in pregnant women and guiding clinical drug selection is a pressing issue.²⁷ The constructed optimal DMPNN-Des model was used to predict the pregnancy-related adverse reaction risks of 22 oral hypoglycemic drugs to guide clinical medication.

The results showed that metformin, rosiglitazone, and chlorpropamide had relatively fewer pregnancy-related adverse reactions, each potentially causing two types of adverse reactions (Figure 5). Furthermore, the prediction results

indicated that saxagliptin and nateglinide had the highest risk, each potentially causing 29 types of adverse reactions (Figure 5). Therefore, the clinical use of these drugs in GDM treatment should be approached with caution.

Analysis of Substructure Alert. At the SOC level, two ring-based substructures, 27 path-based substructures, and one functional group substructure were extracted from the FAERS data set, achieving an average accuracy of 0.8852 ± 0.0681 . For the eight HLGT adverse reaction categories, alert substructures

were successfully extracted with an accuracy of 0.8627 ± 0.0699 . At the PT level, alert substructures for 26 pregnancy-related adverse reactions were identified, with an accuracy of 0.8643 ± 0.0878 (Table S3).

Among these risk substructures from the FAERS data set, 11 substructures were also present in drugs associated with pregnancy-related adverse reactions in the SIDER database and in drugs with fetal malformation risks in the DailyMed database. These substructures include *N,N'*-methylenediacetamide, 1-butene, acetamide, propyne, *N,N*-dimethylacetamide, cyclohexene, terminal alkyne, isopentyl group, 3-hydroxypropynyl, dimethyl ether, and benzyl group (Table 2).

DISCUSSION

This study utilized a large data set from the FAERS database and employed advanced Cheminformatics methods to analyze pregnancy-related ADRs and develop a DMPNN model. Compared to other graph-based DL models (GNN, GCN, and GAT) and traditional descriptor-based ML models (SVM, RF, and XGBoost), the DMPNN model demonstrated superior predictive performance.

Data for Modeling and Methods for Classification of High-Risk Drugs. ADR data can be collected from spontaneous reporting systems (such as the FDA's FAERS, WHO's VigiBase, and EMA's EudraVigilance), EHRs, medical insurance databases (such as Medicare and Medicaid in the U.S.), and clinical trial data.²⁸ For example, Shtar et al.⁵ utilized EHR and cohort study data to develop logistic regression and RF models to predict FDA pregnancy categories for drugs (A, B, C, D, X), assessing potential fetal harm. However, due to the limitations of the available data, their research could target only a specific type of pregnancy-related adverse reaction or focus on a particular drug for study and prediction. In comparison to data from EHRs and expert assessments, our study's data sources offered significant advantages. Our modeling and validation data excelled, including volume, coverage, diversity, representativeness, accessibility, cost, analytical flexibility, and validation. These strengths have significantly improved the reliability and applicability of our AI models in predicting comprehensive pregnancy-related ADRs.

Our findings indicate that the ROR method ($ROR > 1$) for classifying drugs into high-risk and low-risk groups is more effective for AI model development than traditional ADR monitoring methods such as ROR + CI and BCPNN. We extracted data from the FAERS database and assessed traditional adverse reaction signal detection methods, including the BCPNN method with the IC-2SD approach (IC method) and the ROR method (ROR + CI method).⁸ Consistent with previous studies,²⁹ the BCPNN indicator demonstrated high sensitivity and specificity, identifying the fewest drugs flagged as positive (Table 1). However, models constructed using these traditional methods showed suboptimal predictive performance on both internal and external test sets (Figures 3 and 4). When applying the $ROR > 1$ criterion to classify drugs into high-risk and low-risk categories, we found that the high-risk drugs identified included those flagged by traditional methods (Figure S1). Notably, the performance of the model was significantly enhanced, with accuracy on the external test set being markedly higher than that achieved by the other two methods (72.53% vs 49.45% and 27.47%). This increased effectiveness is likely due to the balanced nature of the high-risk and low-risk drug data sets created using the ROR method.

Drug Characteristics and the Methods Used for Modeling. The drug features based on graphs or descriptors in this study are not only easily accessible but can also predict multiple pregnancy-related adverse reactions simultaneously, making the model more broadly applicable while maintaining a comparable prediction accuracy. In contrast, the model developed by Shtar et al.⁵ relies on features such as anatomical therapeutic chemical codes, targets, drug-metabolizing enzymes, and transporters, which are difficult to obtain, especially for new or unapproved drugs.

DMPNN showed excellent predictive performance for both the test set and the external test set. The superior performance of DMPNN can be attributed to its ability to effectively capture complex relationships within molecular graphs. Unlike GCN and GAT, which mainly focus on local neighborhood aggregation, DMPNN is designed for directed graphs and integrates complex molecular information through directed message passing. This enhances its ability to simulate the complex interactions between molecular structure and pharmacological effects, resulting in more accurate predictions.²⁴ The GCN and GAT models, both reported and used in a previous study, exhibit poor predictive efficacy for adverse reactions (AUC: 0.5–0.6),²⁵ while the superior performance of the DMPNN has been demonstrated and applied in multiple studies.^{16,18}

Hierarchy of MedDRA and the Performance of the Model. At the SOC level, the performance of the DMPNN model and other models was lower (AUC: 0.456–0.630), but consistent with reported results for graph-based models. For example, Jiang et al.¹¹ constructed multiple ML models (XGBoost, SVM, RF) and DL models (DNN, GCN, GAT, MPNN) using the SIDER data set at the SOC level to predict the risk of adverse reactions across various SOC categories, with AUCs ranging from 0.602 to 0.665. Similarly, other studies using the SIDER data set reported AUCs of 0.503 ± 0.005 for GCN models and $0.655 \pm 0.033/0.646 \pm 0.016$ for DMPNN models.^{30,31} The lower predictive performance at the SOC level is likely due to the need to predict adverse reactions across a broad range of system organs, each involving complex and diverse biological mechanisms. This complexity limits the model's ability to achieve high accuracy at this level.

However, when the prediction task is focused on the HLGT and PT levels, the model performance significantly improves. These levels more specifically describe particular types of adverse reactions, reducing complexity and uncertainty and enabling the models to more accurately capture relevant molecular features and reaction mechanisms. These findings suggest that the detailed information provided at the HLGT and PT levels can enhance high-precision risk prediction, thereby improving the performance of AI models in predicting ADRs. However, it is noteworthy that external validation revealed low predictive accuracy of this study's model for PT-level adverse reactions in Class 13 and Class 17. The underlying reasons for this remain unclear; however, planned updates to the FAERS and SIDER data sets, including expanding the range of drugs, are expected to enhance the model's predictive accuracy in future applications.

Substructure Alert. The substructures identified in this study are present in many drugs or molecules that are toxic to both the mother and the fetus, further confirming their teratogenic effects during pregnancy. For example, *N,N*-dimethylacetamide and acetamide have shown developmental toxicity in rats and Himalayan rabbits, causing fetal toxicity

such as increased skeletal variations, significant decreases in fetal and placental weight, increased soft tissue and skeletal variations, reduced fetal weight and live births, and higher intrauterine mortality, while also being toxic to the mother.^{32,33}

4-Vinylcyclohexene diepoxide, which contains the cyclohexene structure, can significantly reduce litter sizes and ovarian reserves in mice, increase the incidence of small ear deformities, and cause DNA damage.³⁴ Ethylene glycol dimethyl ether, diethylene glycol dimethyl ether, and triethylene glycol dimethyl ether, which contain the dimethyl ether substructure, have shown dose-dependent maternal and fetal toxicity in rats, causing external, visceral, and skeletal malformations as well as increased maternal liver weight.^{35–37} Benzyl is present in butyl benzyl phthalate and di-*n*-butyl phthalate and their metabolites mono-*n*-butyl and monobenzyl, which exhibit concentration-dependent embryotoxicity and teratogenicity.^{38–42}

Case Study on GDM. GDM is a form of glucose intolerance that occurs during pregnancy, affecting approximately 7.6% of pregnant women in the United States. It is one of the most common obstetric complications.⁴³ GDM can cause both short- and long-term complications for the mother and fetus. Short-term effects include fetal macrosomia (excessive birth weight), hypoglycemia, respiratory distress syndrome, and preterm birth. Long-term effects include an increased risk of type 2 diabetes for both the mother and child.⁴⁴ Additionally, GDM pregnancies can lead to high-risk complications such as preeclampsia (PE) and miscarriage.⁴⁵

The adverse reactions caused by metformin had relatively less impact on the fetus, consistent with the existing research findings. A large trial involving 751 women with GDM compared insulin therapy with metformin (supplemented with insulin if necessary). The perinatal morbidity rates were similar in both groups, including neonatal hypoglycemia, respiratory distress, need for phototherapy, birth trauma, preterm birth, and low Apgar scores.⁴⁶ Another prospective trial indicated that women treated with metformin had lower average blood glucose levels, less weight gain during pregnancy, and a lower incidence of neonatal hypoglycemia compared to those treated with insulin.⁴⁷ A more recent network meta-analysis, including unpublished trials and comparing various treatments, found no significant advantage of metformin over insulin in terms of large-for-gestational-age infants, macrosomia, neonatal hypoglycemia, or cesarean section rates, and no significant difference in preterm birth rates (RR 1.37, 95% CI 0.62–3.01).⁴⁸ Subsequent meta-analyses, including women with type 2 diabetes and GDM, also reported no increased rate of preterm birth.⁴⁹

Additionally, our model predictions identified saxagliptin and nateglinide as having the highest risk. However, there are currently no human studies or related reports, and the FDA does not recommend their use in pregnant women. Due to the extreme lack of clinical data, this prediction cannot be confirmed.

Limitations. Despite the promising results, this study has several limitations. First, relying on voluntarily reported FAERS data may introduce bias and underreporting. While these limitations are inherent to postmarketing surveillance databases, we have taken several measures to mitigate these effects. Specifically, we have used statistical methods to handle missing or incomplete data and we employed robust filtering techniques to exclude low-quality reports. Prior to calculating the ROR using FAERS data, we implemented a systematic,

standardized deduplication process, retaining only the latest reports. This approach was designed to improve the data quality and accuracy in the database, thereby minimizing potential biases. Second, although the DMPNN model demonstrated strong performance, its generalizability needs further validation with larger and more diverse data sets. Integrating other data sources, such as EHRs and genomic data, could enhance the model's predictive ability. Future research should focus on expanding data sets and integrating multiomics data to provide a more comprehensive view of drug safety. Additionally, exploring other neural network architectures and ensemble methods could further improve the prediction accuracy. Combining cheminformatics with fields such as clinical individual information, pharmacogenomics, and systems biology could enhance our understanding of drug safety and efficacy.

CONCLUSIONS

In conclusion, this study demonstrated the potential of the ROR method in identifying high-risk drugs for pregnancy-related adverse reactions by employing a DMPNN model that integrates molecular graphs and descriptors. The DMPNN model, validated with external data sets, proved to be a robust and reliable tool for assessing drug safety. This approach facilitates the development of safer drugs for use during pregnancy and their clinical applications, ultimately contributing to improved patient treatment outcomes.

ASSOCIATED CONTENT

Data Availability Statement

Additional data have been made available in the electronic Supporting Information.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.4c01657>.

Venn diagrams depicting overlap of low-risk and high-risk drugs at various classification levels using different risk assessment metrics; drugs and pregnancy-related adverse reactions extracted from the FAERS database; model metrics with different methods; and substructure alert extracted from the FAERS data set (PDF)

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Author Contributions

JFP, GPY, and DSC designed the research; JFP and LF performed the research; JFP analyzed the data; JFP wrote the manuscript and GPY and DSC revised it critically; GPY and DSC acted as actively involved advisors throughout the project. All authors read and approved the final manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work supported by youth fund of National Natural Science Foundation of China (grant number: 81603192), Natural Science Foundation of Hunan Province (grant number: 2023JJ40900).

ABBREVIATIONS

ACC	accuracy
ADRs	adverse drug reactions
AUC	area under the curve
BCPNN	Bayesian confidence propagation neural network
CI	confidence interval
DGL	deep graph library
DL	deep learning
DMPNN	directed message passing neural networks
ECFP4	extended connectivity fingerprint
EHRs	electronic health records
FAERS	FDA adverse event reporting system
FDA	Food and Drug Administration
FFN	feed-forward neural network
GAT	graph attention networks
GCN	graph convolutional networks
GDM	gestational diabetes mellitus
GNN	graph neural networks
HLGT	high-level group term
IC	information component
MACCS	molecular access system
ML	machine learning
PT	preferred term
RBF	radial basis function
RF	random forest
ROR	reporting odds ratio
SE	sensitivity
SMILES	simplified molecular input line entry system
SOC	system organ class
SP	specificity
SVM	support vector machine

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NOTE ADDED AFTER ISSUE PUBLICATION

Author name Dongshen Cao was corrected to Dongsheng Cao on December 30, 2024.